

Project Design Phase

Problem – Solution Fit Template

Date	30 September 2026
Team ID	SWTID-2026-7612
Project Name	Import Data Using Transform Maps (Spreadsheet)
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- Solve complex problems in a way that fits the state of your customers.
- Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- Sharpen your communication and marketing strategy with the right triggers and messaging.
- Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- Understand the existing situation in order to improve it for your target group.

Template:

No.	Problem – Solution Fit Area	Project-specific content
1	Customer Segment(s) (CS)	HR personnel / HR managers who maintain employee information and need a centralized way to import and manage employee records.
2	Jobs-to-be-Done / Problems (J&P)	Import employee information from spreadsheets into ServiceNow accurately, avoid unnecessary duplicate records, and make the resulting data available for reporting.
3	Triggers (TR)	The need to move employee data into a centralized

		system, reduce manual data entry, update existing employee records, and prepare employee information for reporting.
4	Emotions: Before / After (EM)	Before: difficulty in maintaining and checking spreadsheet-based employee information. After: employee data is available in a structured ServiceNow table and can be viewed through reports and dashboards.
5	Available Solutions (AS)	Spreadsheets and manual record entry are available alternatives. These can require repeated manual work and make centralized reporting more difficult.
6	Customer Constraints (CC)	Employee data must follow a consistent structure; existing employee records should not be unnecessarily duplicated; the process should be usable within the organization's existing ServiceNow environment.
7	Behaviour (BE)	HR users prepare employee data in a spreadsheet, load it into ServiceNow, review the staged records, run the Transform Map, and verify the resulting employee records.
8	Channels of Behaviour (CH)	Online: ServiceNow web interface, Import Sets, Transform Map configuration and Employee Analytics Dashboard. Offline: preparation and review of employee information before importing the spreadsheet.
9	Problem Root Cause (RC)	Employee information maintained separately in spreadsheets requires manual handling and makes it harder to maintain a consistent centralized employee dataset for reporting.
10	Your Solution (SL)	A ServiceNow-based import

		workflow using a spreadsheet, Import Set staging table and Transform Map. Employee ID is configured as a coalesce field to match existing records, and the resulting Employee Test data is used for analytics.
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References:

<https://www.ideahackers.network/problem-solution-fit-canvas/>

<https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>

Project Design Phase

Proposed Solution Template

Date	30 September 2026
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Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Employee information is often prepared and maintained in spreadsheets, making centralized management, accurate updating and reporting difficult. A structured method is required to import employee data into ServiceNow and manage it consistently.
2	Idea / Solution description	Use ServiceNow Import Sets and a Transform Map to import employee data from a spreadsheet. The spreadsheet is loaded into the Employee Import staging table, mapped to the Employee Test table, and transformed into employee records. Employee ID is used as the coalesce field for matching existing records. The resulting data is presented through the Employee Analytics Dashboard.
3	Novelty / Uniqueness	The solution combines spreadsheet-based bulk import with ServiceNow transformation and employee ID-based record matching, reducing repetitive manual data entry while keeping the employee

		information within the ServiceNow platform.
4	Social Impact / Customer Satisfaction	HR users can spend less time entering employee records manually and can access a structured employee dataset for reporting. Clearer employee information can support faster review and easier organizational reporting.
5	Business Model (Revenue Model)	The solution is intended as an internal enterprise ServiceNow capability. It does not require a separate direct revenue model; value is obtained through improved data management efficiency and reduced manual effort.
6	Scalability of the Solution	The same Import Set and Transform Map workflow can process additional spreadsheet records without changing the basic architecture. Employee ID matching also supports maintaining existing records as the dataset grows.

Project Design Phase

Solution Architecture

Date	30 September 2026
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Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

Solution Architecture Diagram – Import Data Using Transform Maps (Spreadsheet)

HR User	→	Spreadsheet	→	ServiceNow Import Set
			↓	
	Employee Import Staging Table	→	Transform Map	
			↓	
	Employee Test Table	→	Employee Analytics Dashboard	

Figure 1: Architecture and data flow of the Import Data Using Transform Maps (Spreadsheet) solution.

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>

Project Design Phase-II

Customer Journey Map

Scenario

Importing and managing employee information in ServiceNow using a spreadsheet and Transform Map.

Steps

Entice	Enter	Engage	Exit	Extend
HR identifies the need to maintain employee information in a centralized system and prepares the spreadsheet.	HR opens ServiceNow and loads the spreadsheet through the Import Set process.	The staged records are transformed using the Transform Map and mapped to the Employee Test table.	HR reviews the imported employee records and verifies the employee details.	HR uses the Employee Analytics Dashboard to view employee information by department and location and can use the updated data for reporting.

Interactions

Steps	What does the person typically experience?	What interactions do they have at each step along the way?
Entice	The HR user decides to move employee data from a spreadsheet into ServiceNow.	Reviews the employee information and prepares the spreadsheet with the required fields.
Enter	The HR user begins the import process.	Uploads the spreadsheet through ServiceNow Import Sets and checks the staging data.
Engage	The employee data is processed and transferred into the target table.	Runs the Transform Map, which maps the source fields to the Employee Test table and uses Employee ID for matching.
Exit	The HR user verifies the imported records.	Checks employee name, email, department, employee ID and location in the Employee Test list.
Extend	The HR user uses the stored data for reporting.	Views the Employee Analytics Dashboard,

		including employee distribution by department and location.
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Project Design Phase-II

Data Flow Diagram & User Stories

Date	30 September 2026
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Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: (Simplified)

Data Flow Diagram – Import Data Using Transform Maps

Source	Process	Data Store / Target	Output
HR User Spreadsheet	Import Set Loads spreadsheet records	Employee Import (Staging Table)	Staged employee data
Staged employee data	Transform Map Field mapping + transformation Employee ID used for coalesce	Employee Test Table	Employee records
Employee Test Table	Reporting / Dashboard	Employee Analytics Dashboard	Employee information by Department and Location

Data flow: HR User Spreadsheet → Import Set / Employee Import staging table → Transform Map → Employee Test table → Employee Analytics Dashboard.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
HR User	Import	USN-1	As an HR	The	High	Sprint-1

	Employee Data		user, I can import employee information from a spreadsheet into ServiceNow.	spreadsheet records are loaded into the Import Set staging table.		
HR User	Transform Employee Data	USN-2	As an HR user, I can transform the imported spreadsheet data into the Employee Test table.	The Transform Map maps the spreadsheet fields to the correct target fields.	High	Sprint-1
HR User	Employee Identification	USN-3	As an HR user, I can use Employee ID to match existing employee records during transformation.	Records with an existing Employee ID are matched rather than unnecessarily duplicated.	High	Sprint-1
HR User	Employee Record Review	USN-4	As an HR user, I can view the imported employee details in the Employee Test table.	Employee name, email, department, employee ID and location are displayed correctly.	High	Sprint-1
HR User	Employee Analytics	USN-5	As an HR user, I can view employee information through the Employee Analytics Dashboard.	The dashboard displays employee information by department and location.	Medium	Sprint-2

Administrator	Configuration Management	USN-6	As an administrator, I can maintain the Import Set and Transform Map configuration.	The import and transformation configuration can be maintained when required.	Medium	Sprint-2
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Project Design Phase-II

Solution Requirements (Functional & Non-functional)

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Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Import Employee Data	Import employee information from a spreadsheet using ServiceNow Import Sets.
FR-2	Transform Employee Data	Map spreadsheet columns to the corresponding Employee Test table fields using a Transform Map.
FR-3	Employee Identification	Use Employee ID as the coalesce field to identify existing employee records during transformation.
FR-4	Employee Record Management	Create new employee records and update matching records through the transformation process.
FR-5	Employee Information	View imported employee information including employee name, email, department, employee ID and location.
FR-6	Analytics	Display employee information by department and location through the Employee Analytics Dashboard.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The import and employee record management

		process should be simple for HR users to understand and use.
NFR-2	Security	Access to employee information should be controlled using ServiceNow roles and ACLs.
NFR-3	Reliability	Valid spreadsheet records should be processed consistently and transformation results should be available for verification.
NFR-4	Performance	Bulk spreadsheet records should be processed through Import Sets and Transform Maps without manual entry for each record.
NFR-5	Availability	Employee records and dashboards should be available to authorized users through ServiceNow.
NFR-6	Scalability	The Import Set and Transform Map approach should support additional employee records without changing the basic data flow.

Project Design Phase-II

Technology Stack (Architecture & Stack)

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Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

Architecture - Import Data Using Transform Maps

Layer / Component	Flow
Data Source	Spreadsheet containing employee information
Import Layer	ServiceNow Import Set loads records into the Employee Import staging table
Transformation Layer	Sample Spreadsheet Import Transform Map maps source fields to target fields
Data Layer	Employee Test table stores the transformed employee records
Reporting Layer	Employee Analytics Dashboard presents employee information by department and location

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	ServiceNow Web UI
2	Application Logic-1	Logic for a process in the application	ServiceNow Import Sets
3	Application Logic-2	Logic for a process in the application	ServiceNow Transform Maps
4	Application Logic-3	Logic for a process in the application	Transform Map Coalesce using

			Employee ID
5	Database	Data Type, Configurations etc.	ServiceNow platform database
6	Cloud Database	Database Service on Cloud	ServiceNow-managed cloud database
7	File Storage	File storage requirements	Spreadsheet / Local Filesystem
8	External API-1	Purpose of External API used in the application	Not used
9	External API-2	Purpose of External API used in the application	Not used
10	Machine Learning Model	Purpose of Machine Learning Model	Not used
11	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration:	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	List the open-source frameworks used	No separate open-source framework; ServiceNow platform
2	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	ServiceNow Roles and ACLs
3	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Import Sets and Transform Maps support batch employee-data processing
4	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	ServiceNow Cloud Platform
5	Performance	Design consideration for the performance of the application (number of requests per sec,	Bulk import and transformation reduce manual record-by-record data entry

		use of Cache, use of CDN's) etc.	
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References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>

1. Empathize & Discover

Empathy Map Canvas

Date	30 September 2026
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Maximum Marks	4 Marks

Template description: An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes. It is a useful tool to help teams better understand their users. Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Reference: <https://www.mural.co/templates/empathy-map-canvas>

Project context: The empathy map focuses on HR personnel/HR managers who need to maintain employee skill and training information securely, and employees whose records must be accurately represented.

SAYS	"I need employee training/skill information in one place." "I need the employee's department and other details to be available for reporting." "I should be able to import spreadsheet data without manually entering every record." "Only authorized HR users should be able to access or modify these records."
THINKS	"Manual data entry can cause errors and duplicate records." "If employee information changes, the system should update the correct record." "Reports should help me understand employees by department and location." "Employee information must not be exposed to unauthorized users."
DOES	Collects employee information in spreadsheets. Imports data into the ServiceNow system. Reviews and updates employee records. Uses reports/dashboards to analyse employees by location and department. Relies on employee ID to identify an existing employee record during import.
FEELS	Concerned about inaccurate or outdated employee records. Concerned about privacy and unauthorized access.

	More confident when data imports complete successfully and records are easy to report on.
GOALS	Maintain a centralized employee skill/training record. Reduce repetitive manual data entry. Match imported data to the correct employee using Employee ID. Make department/location information available for reporting. Restrict access to authorized HR users.
PAINS / CHALLENGES	Scattered spreadsheet-based records. Manual updates can be time-consuming and error-prone. Duplicate records may be created if existing employees are not matched correctly. Limited visibility for reporting and analytics. Security and access control must be enforced.

Empathy Map – Key Insight

The user needs a centralized and secure ServiceNow-based Skill Wallet that can accept structured spreadsheet data, identify employees reliably, support reporting, and restrict sensitive employee information to authorized users. These needs directly informed the selected solution and later development activities.

2. Brainstorm & Idea Prioritization

Brainstorm & Idea Prioritization Template

Date	30 September 2026
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Template description: Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you are not sitting in the same room.

Reference: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>

Step-1: Team Gathering, Collaboration and Select the Problem Statement

The team reviewed the difficulties involved in maintaining employee skill/training information using spreadsheets and identified the need for a centralized ServiceNow solution. The selected problem statement was:

Selected Problem Statement: HR personnel need a secure and centralized way to import, maintain and analyse employee skill/training information so that records remain accurate, useful for reporting, and accessible only to authorized users.

Step-2: Brainstorm, Idea Listing and Grouping

Idea Group	Brainstormed Ideas	Purpose / Expected Benefit
Data Management	Create a centralized employee/skill record table; import spreadsheet data using Import Sets; use a Transform Map to map source fields to target fields.	Reduce manual entry and keep records structured in ServiceNow.
Record Matching	Use Employee ID as the matching/coalesce field during transformation.	Update the correct existing employee instead of creating unnecessary duplicate records.
Information Enrichment	Maintain department and location information with employee records; use reference/dot-walking where applicable.	Provide additional context for users and reporting.
Analytics & Reporting	Create employee list reports; analyse employees by location and department; provide a dashboard.	Make workforce information easier to understand and monitor.
Security & Access Control	Create HR-specific roles and ACLs for read/write access.	Protect employee information and limit access to authorized HR users.
Usability / Navigation	Expose the relevant table and modules through the ServiceNow application/menu.	Make the solution easier for intended users to access.

Step-3: Idea Prioritization

Ideas were considered against practical project needs: relevance to the identified problem, implementation feasibility in ServiceNow, usefulness to HR, data accuracy, reporting value, and security. The selected concept combines the core ideas rather than treating them as isolated features.

Solution Component	Problem Addressed	Implementation Evidence	Role in Selected Solution
Spreadsheet Import	Manual data entry / fragmented records	Import Set + staging table	Core data ingestion
Transform Map + Employee ID Coalesce	Incorrect matching / duplicates	Field mappings and Employee ID coalesce	Accurate insert/update handling
Reports & Dashboard	Poor visibility	Employee list, location chart, department chart	Analytics and decision support
Roles & ACLs	Unauthorized access	HR role and access rules	Security and controlled access

Selected Solution: A ServiceNow Skill Wallet that centralizes employee skill/training-related information, supports spreadsheet-based data ingestion and employee matching, provides reporting/analytics, and applies role-based access controls.

3. Define the Problem Statements

Date	30 September 2026
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Maximum Marks	2 Marks

Template description: Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love. A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

Reference: <https://miro.com/templates/customer-problem-statement/>

Customer Problem Statement Template

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	an HR manager responsible for employee skill and training records	maintain accurate employee information in one centralized system	the information is often maintained or exchanged through spreadsheets and manual processes	manual handling makes it harder to keep records consistent, identify the correct employee, and maintain current information	concerned about errors, duplication, and time spent maintaining records
PS-2	an HR manager who needs employee information for reporting	import employee data and analyse it by department and location	raw employee data does not automatically provide a clear, secure reporting view	the team needs structured ServiceNow records, reliable matching, useful reports, and controlled access	concerned that useful information may be difficult to analyse or may be exposed to unauthorized users

Problem Definition

HR teams need a centralized ServiceNow Skill Wallet to manage employee skill/training information securely and consistently. The solution should support spreadsheet imports, accurate employee identification through Employee ID matching, structured employee information, analytics by department and location, and controlled access for authorized users.

Project Planning — Planning Logic

Project: Import Data using Transform Maps (Spreadsheet)

Agile Terminology

A Sprint is a fixed period or duration in which a team works to complete a set of tasks.

An Epic is a big task or project that is too large to complete in one sprint. It is broken down into smaller tasks (stories) that can be completed over multiple sprints.

A Story is a small task. It is part of an Epic.

A Story Point is a number that represents how much effort a story takes to complete (usually in the form of the Fibonacci series).

Story Point Scale (1, 2, 3, 5):

- 1 — Very Easy task
- 2 — Normal task
- 3 — Moderate task
- 5 — Difficult task

Sprint 1 — Data Setup, Import & Transform

Data Collection & Table Creation (Epic 1)

Creating Spreadsheet (USN1) — 2

Creating Employee Table (USN2) — 2

Import Set & Transform Map (Epic 2)

Creating Import Set Table (USN3) — 3

Creating Transform Map (USN4) — 3

Transform & Validate (Epic 3)

Transform Data & Validate (USN5) — 2

Total Story Point in Sprint 1 = 2 + 2 + 3 + 3 + 2 = 12

Sprint 2 — Deduplication, Reporting & Dashboard

Data Integrity – Coalesce (Epic 4)

Enable Coalesce to Avoid Duplicate Records (USN6) — 3

Inserting New Data in Excel Format (USN7) — 2

Reports (Epic 5)

Employee List Report (USN8) — 2

Employees by Location – Bar Chart (USN9) — 2

Employees by Department – Pie Chart (USN10) — 2

Dashboard (Epic 6)

Adding Reports to Dashboard (USN11) — 5

Total Story Point in Sprint 2 = 3 + 2 + 2 + 2 + 2 + 5 = 16

Total Story Points

Sprint 1 = 12

Sprint 2 = 16

Velocity Calculation

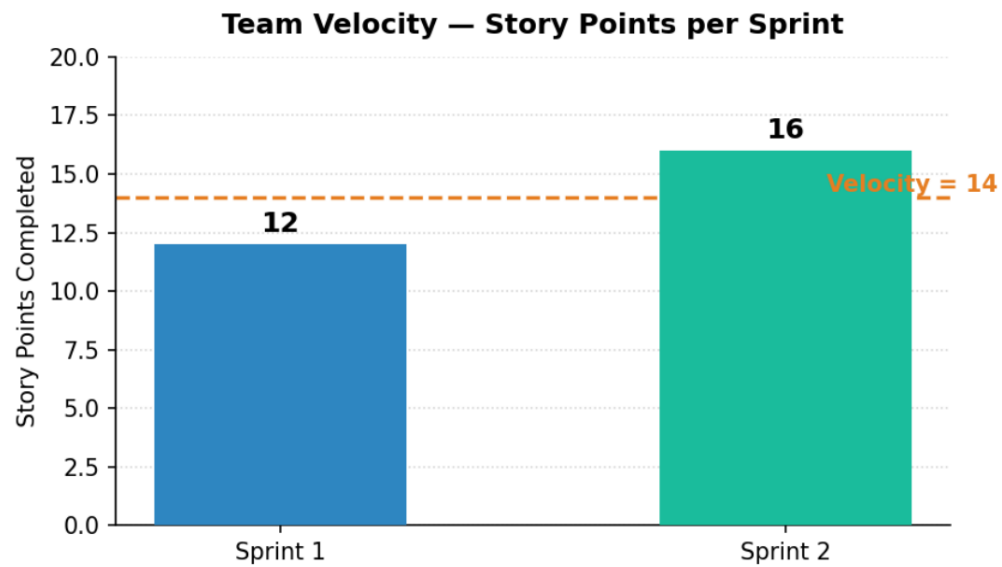
Velocity = Total Story Points Completed / Number of Sprints

Total Story Points = 12 + 16 = 28

Number of Sprints = 2

Velocity = 28 / 2 = 14 (Story Points per Sprint)

Your team's velocity is 14 Story Points per Sprint.



Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story Points)

Date	29 September 2026
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Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection & Table Creation	USN-1	As an admin, I can create a spreadsheet of employee details (ID, Name, Email, Department, Location).	2	High	
Sprint-1		USN-2	As an admin, I can create the target Employee table in ServiceNow.	2	High	
Sprint-1	Import Set & Transform Map	USN-3	As an admin, I can create an Import Set table to stage the spreadsheet data.	3	High	
Sprint-1		USN-4	As an admin, I can create a Transform Map to map import fields to the Employee table.	3	High	
Sprint-1	Transform & Validate	USN-5	As an admin, I can transform and validate the imported employee data.	2	Medium	
Sprint-2	Data Integrity (Coalesce)	USN-6	As an admin, I can enable Coalesce on Employee ID to avoid duplicate records.	3	High	
Sprint-2		USN-7	As an admin, I can insert/update employee data from an Excel file through the transform.	2	Medium	
Sprint-2	Reports	USN-8	As a user, I can view an Employee List report.	2	Medium	
Sprint-2		USN-9	As a user, I can view Employees by Location as a bar chart.	2	Low	
Sprint-2		USN-10	As a user, I can view Employees by Department as a pie chart.	2	Low	
Sprint-2	Dashboard	USN-11	As a user, I can view all reports together on a single dashboard.	5	High	

Sprint-1 total = 2 + 2 + 3 + 3 + 2 = 12 | **Sprint-2 total = 3 + 2 + 2 + 2 + 2 + 5 = 16**

Project Tracker, Velocity & Burndown Chart (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (Planned End Date)	Sprint Release Date (Actual)
Sprint-1	12	6 Days	22 Sep 2026	27 Sep 2026	12	27 Sep 2026
Sprint-2	16	6 Days	29 Sep 2026	04 Oct 2026	16	04 Oct 2026

Velocity:

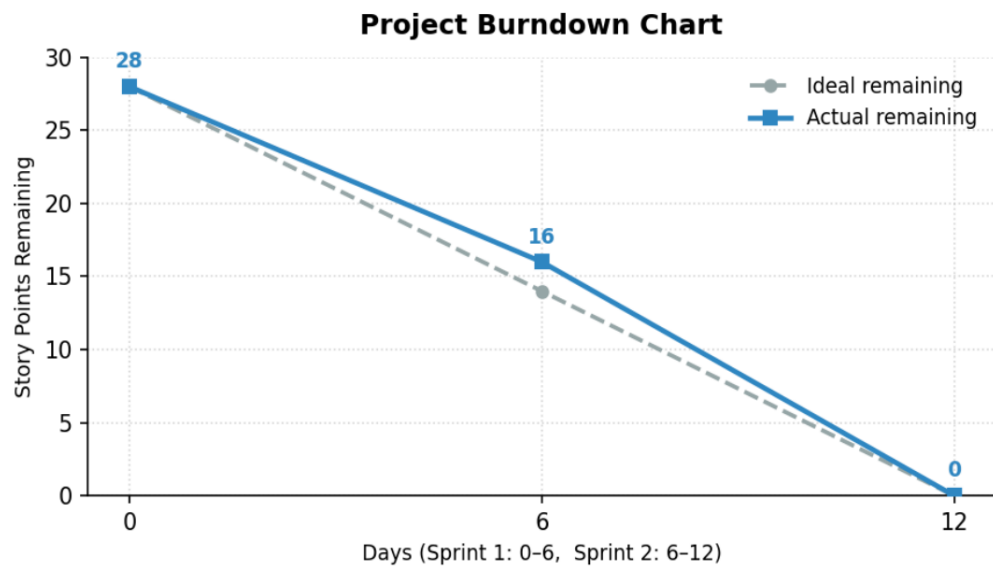
Our sprint duration is 6 days and the team's velocity is 14 story points per sprint. The average velocity (AV) per day is the story points delivered per working day:

$$AV = \text{velocity} \div \text{sprint duration} = 14 \div 6 \approx 2.3 \text{ story points per day}$$

Note: velocity per day = velocity ÷ sprint duration (story points delivered per day). In the original sample the label read "sprint duration ÷ velocity" but the numbers used velocity ÷ duration (20 ÷ 10 = 2), so the formula above is the corrected version.

Burndown Chart:

A burndown chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum, but can be applied to any project containing measurable progress over time. The chart below tracks the remaining story points across our two sprints against the ideal burndown line.



Reference:

<https://www.atlassian.com/agile/project-management>
<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>
<https://www.atlassian.com/agile/tutorials/epics>
<https://www.atlassian.com/agile/tutorials/sprints>
<https://www.atlassian.com/agile/project-management/estimation>
<https://www.atlassian.com/agile/tutorials/burndown-charts>

Project Development Phase
Model Performance Test

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Maximum Marks	

Model Performance Testing:

S.No.	Parameter	Values	Screenshot
1	Model Summary	Not Applicable — this project does not build or train any AI/ML model.	
2	Accuracy	Training Accuracy — N/A Validation Accuracy — N/A	
3	Fine Tuning Result (if Done)	Validation Accuracy — N/A	

Note: The “Import Data using Transform Maps (Spreadsheet)” project is a ServiceNow data-import and reporting solution built with Import Sets, Transform Maps, Coalesce (to avoid duplicate records), and Reports & Dashboards. It does not involve any Artificial Intelligence or Machine Learning model, so model summary, accuracy, and fine-tuning metrics are not applicable to this project.

Functional & Performance Testing Template

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Maximum Marks	

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Employee data field validation	Enter valid and invalid values in Employee ID, Name, Email, Department, Location	Valid rows accepted; rows with missing required fields are flagged	Valid rows imported into Employee Test; missing name values identified and corrected	Pass
FT-02	Spreadsheet import to Import Set	Load the .xlsx into the Employee Import staging table (Sheet 1, Header row 1)	All rows staged in the Import Set table	Rows loaded into u_employee_import successfully	Pass
FT-03	Transform Map field mapping	Run the transform and verify Import Set fields map to the Employee Test columns	All fields mapped correctly to the target table	All fields mapped; employee names populated after correcting u_name → u_employee_name	Pass
FT-04	Coalesce – duplicate prevention	Re-import records that use existing Employee IDs	Existing records updated; no duplicate rows created	4 records re-imported → 4 updated, 0 inserted (no duplicates)	Pass
PT-01	Transform run-time test	Time the transform execution for a data load	Transform completes within a few seconds	Completed in 0 seconds – State: Complete / Success	Pass
PT-02	Bulk import test	Import the full employee dataset (17 records) in one load	All records processed without error	17 records processed; State Complete; 0 errors	Pass
PT-03	Re-import load test	Load the same file again through the same transform map	No crash; duplicates ignored via coalesce	Re-run → 4 ignored, 0 inserted, 0 updated; no errors	Pass

Note: FT = Functional Test, PT = Performance Test. Scenarios are adapted to this ServiceNow data-import project (Import Sets, Transform Map, Coalesce, Reports & Dashboards); the Actual Results reflect the outcomes observed during testing.

Project Development Phase
Model Performance Test

Date	30 September 2026
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Maximum Marks	10 Marks

Model Performance Testing:

S.No.	Parameter	Values	Screenshot
1	Metrics	Regression Model: MAE – N/A, MSE – N/A, RMSE – N/A, R ² score – N/A Classification Model: Confusion Matrix – N/A, Accuracy Score – N/A, Classification Report – N/A	
2	Tune the Model	Hyperparameter Tuning – N/A Validation Method – N/A	

Note: The “Import Data using Transform Maps (Spreadsheet)” project is a ServiceNow data-import and reporting solution (Import Sets, Transform Maps, Coalesce, Reports & Dashboards). It does not build or train any machine-learning model, so regression/classification metrics and model tuning are not applicable to this project.

Project Development Phase
Model Performance Test

Date	30 September 2026
Team ID	SWTID-2026-7612
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	

Model Performance Testing:

S.No.	Parameter	Screenshot / Values
1	Data Rendered	17 employee records rendered in ServiceNow reports and dashboard (Employee ID, Name, Email, Department, Location).
2	Data Preprocessing	Data staged via Import Set, mapped through the Transform Map, and deduplicated using Coalesce on Employee ID before reporting.
3	Utilization of Data Filters	Reports grouped/filtered by Location and Department; Employee List sorted by Employee ID.
4	DAX Queries Used	N/A — DAX is Power BI-specific. ServiceNow reporting uses Group By + Count aggregation instead of DAX.
5	Dashboard design	Employee Analytics Dashboard. No. of Visualizations / Graphs – 3 (Employee List report, Employees by Location bar chart, Employees by Department pie chart).
6	Report Design	No. of Visualizations / Graphs – 3 Employee List report, Employees by Location (bar), Employees by Department (pie).

Note: This project used ServiceNow's native reporting and dashboards (Platform Analytics), not Power BI. The parameters above are filled with the equivalent ServiceNow reporting work; DAX Queries are Power BI-specific and do not apply.

Project Development Phase Model Performance Test

Date	30 September 2026
Team ID	SWTID-2026-7612
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	

Model Performance Testing:

S.No.	Parameter	Values	Screenshot
1	Model / Setup Summary	ServiceNow setup for employee data management using a Table (Employee Test), Fields, a Transform Map and Reports. Note: Records are imported via an Import Set + Transform Map. If the Employee ID matches an existing record it is updated (Coalesce); a new ID creates a new record; invalid or mismatched rows are skipped/errored.	Refer to Fig. 2 (below)
2	Accuracy	Training Accuracy – N/A, Validation Accuracy – N/A (no ML model). Data import accuracy: 17/17 records imported successfully, 0 errors (100%).	Refer to Fig. 1 (below)
3	Confidence Score (Only YOLO Projects)	N/A — this is not a YOLO / object-detection project, so no confidence score applies.	

Evidence Screenshot

The screenshot shows the ServiceNow 'Transform Histories' page. A table lists the history of transform jobs. The first entry is highlighted, showing a successful import of 4 records with 0 errors.

Started	State	Completed	Run time	Set	Import set table	Total	Inserts	Updates	Ignored	Skipped	Errors	Transform Map
2026-09-29 01:21:50	Complete	2026-09-29 01:21:50	0 Seconds	ISET0010007	Employee Import [c_employee_import]	4	0	4	0	0	0	Sample Spreadsheet Import

Fig. 1 — Transform completed successfully: State “Complete”, 0 errors (Sample Spreadsheet Import transform map).

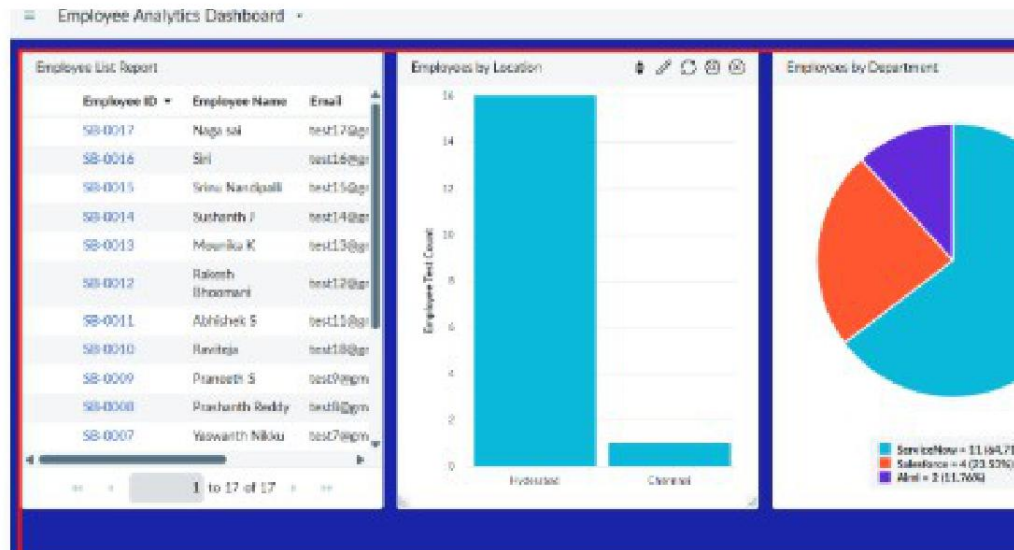


Fig. 2 — Employee data rendered in the ServiceNow dashboard: Employee List, Employees by Location (bar chart), and Employees by Department (pie chart).

Note: This project was built on ServiceNow (not Salesforce); the data-management setup above is the ServiceNow equivalent (Tables, Fields, Transform Map, Reports). Training/validation accuracy and YOLO confidence score relate to ML models and do not apply.

Project Development Phase Performance Test

Date	30 September 2026
Team ID	SWTID-2026-7612
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	

Model Performance Testing:

S.No.	Parameter	Screenshot / Values
1	Data Rendered	17 employee records rendered in ServiceNow reports and dashboard (Employee ID, Name, Email, Department, Location).
2	Data Preprocessing	Data staged via Import Set, mapped through the Transform Map, and deduplicated using Coalesce on Employee ID before reporting.
3	Utilization of Filters	Reports grouped/filtered by Location and Department; Employee List sorted by Employee ID.
4	Calculation Fields Used	N/A — calculated fields are Tableau-specific. ServiceNow reporting uses Group By + Count aggregation instead.
5	Dashboard design	Employee Analytics Dashboard. No. of Visualizations / Graphs – 3 (Employee List report, Employees by Location bar chart, Employees by Department pie chart).
6	Story Design	N/A — the Tableau “Story” feature was not used. ServiceNow has no equivalent; the consolidated view is the single Employee Analytics Dashboard (3 visualizations).

Note: This project used ServiceNow's native reporting and dashboards (Platform Analytics), not Tableau. The parameters above are filled with the equivalent ServiceNow reporting work; calculated fields and the Story feature are Tableau-specific and do not apply.

User Acceptance Testing (UAT) Template

Date	30 September 2026
Team ID	SWTID-2026-7612
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	

Project Overview

Project Name: Import Data using Transform Maps (Spreadsheet)

Project Description: A ServiceNow solution that imports employee records from a spreadsheet into ServiceNow using Import Sets and a Transform Map, uses Coalesce on Employee ID to prevent duplicate records, and presents the data through reports and a dashboard.

Project Version: 1.0

Testing Period: 22 September 2026 to 04 October 2026

Testing Scope

Features and functionalities tested:

- Spreadsheet creation and Employee table setup
- Import Set table load from the spreadsheet
- Transform Map field mapping (Employee ID, Name, Email, Department, Location)
- Data transform, validation, and Coalesce (duplicate prevention)
- Inserting / updating new data from Excel
- Reports (Employee List, Employees by Location, Employees by Department) and the Employee Analytics Dashboard

User stories tested:

- USN-1 to USN-11 (see Project Planning backlog)

Testing Environment

URL / Location: ServiceNow Personal Developer Instance (dev instance)

Credentials (if required): Admin access on the developer instance (not disclosed in this document)

Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	Import employee spreadsheet	Load the .xlsx into Employee Import and run the transform	All records imported into the Employee Test table	17 records imported, 0 errors	Pass
TC-002	Field mapping validation	Verify Employee ID, Name, Email, Department, Location map correctly	All fields populated correctly in target table	All fields mapped; names populated	Pass
TC-003	Coalesce duplicate prevention	Re-import records that use existing Employee IDs	Existing records updated; no duplicates created	4 updated, 0 inserted (no duplicates)	Pass
TC-004	Re-import identical data	Load the same file again through the transform	Rows ignored; no changes made	4 ignored, 0 inserted/updated	Pass
TC-005	Employee List report	Open the Employee List report	All employees listed with details	17 employees listed correctly	Pass
TC-006	Employees by Location (bar)	Open the Employees by Location report	Bar chart shows employees per location	Bar chart: Hyderabad 16, Chennai 1	Pass
TC-007	Employees by Department (pie)	Open the Employees by Department report	Pie chart shows department distribution	Pie: ServiceNow 11, Salesforce 4, AIML 2	Pass
TC-008	Employee Analytics Dashboard	Open the dashboard	All three reports display together	Dashboard shows list + bar + pie	Pass

Bug Tracking

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
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BG-001	Duplicate employee records created on re-import	Import data before Coalesce is enabled, then re-import	Medium	Closed	Fixed by enabling Coalesce (true) on Employee ID; existing duplicate rows removed
BG-002	Employee Name field blank for imported records	Import the original spreadsheet and open the Employee Test table	Medium	Closed	Transform map corrected (u_name → u_employee_name); names re-populated for all 17 records
BG-003	Employees by Location shown as pie instead of bar	Open the Employees by Location report	Low	Closed	Report type changed from Pie to Bar

Sign-off

Tester Name: Gopika Sri

Date: 04 October 2026

Signature: _____

Acceptance Testing

UAT Execution & Report Submission

Date	30 September 2026
Team ID	SWTID-2026-7612
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

1. Purpose of Document

The purpose of this document is to briefly explain the test coverage and open issues of the Import Data using Transform Maps (Spreadsheet) project at the time of its release to User Acceptance Testing (UAT). All identified defects were resolved and all test cases passed; there are no open issues at release.

2. Defect Analysis

This report shows the number of resolved or closed bugs at each severity level, and how they were resolved.

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	0	0	0	0	0
Duplicate	0	0	0	0	0
External	0	0	0	0	0
Fixed	0	2	1	0	3
Not Reproduced	0	0	0	0	0
Skipped	0	0	0	0	0
Won't Fix	0	0	0	0	0
Totals	0	2	1	0	3

3. Test Case Analysis

This report shows the number of test cases that have passed, failed, and remained untested.

Section	Total Cases	Not Tested	Fail	Pass
Data Import & Transform	2	0	0	2
Coalesce / Deduplication	2	0	0	2
Reports (List, Location, Department)	3	0	0	3
Dashboard	1	0	0	1
Totals	8	0	0	8

PROJECT REPORT

Import Data Using Transform Maps (Spreadsheet)

Team ID: SWTID-2026-7612

1. INTRODUCTION

1.1 Project Overview

The project, Import Data Using Transform Maps (Spreadsheet), is a ServiceNow-based solution for centralizing employee information imported from spreadsheets. The solution uses an Import Set and staging table to receive spreadsheet records, a Transform Map to move the data into the Employee Test table, and Employee ID as the coalesce field to match existing records. The processed information is then presented through reports and an Employee Analytics Dashboard.

1.2 Purpose

The purpose is to reduce repetitive manual data entry, maintain structured employee records, prevent unnecessary duplicate records, support reporting by department and location, and provide controlled access to employee information through ServiceNow roles and ACLs.

2. IDEATION PHASE

2.1 Problem Statement

HR teams need a centralized ServiceNow Skill Wallet to manage employee skill and training information securely and consistently. The solution should support spreadsheet imports, accurate employee identification through Employee ID matching, structured employee information, analytics by department and location, and controlled access for authorized users.

2.2 Empathy Map Canvas

Area	Key Points
Says	Need employee information in one place; import spreadsheet data without entering every record manually; only authorized HR users should access or modify records.
Thinks	Manual entry can cause errors and duplicates; changed information should update the correct employee; reports should support department and location analysis.
Does	Collects employee information in spreadsheets, imports it into ServiceNow, reviews records, and uses reports/dashboard.
Feels	Concerned about inaccurate records, duplication, privacy and unauthorized access; more confident when imports complete successfully.
Goals	Centralize employee information, reduce manual effort, match employees using Employee ID, support reporting, and control access.

Pains / Challenges	Scattered spreadsheets, time-consuming manual updates, duplicate records, limited reporting visibility, and access-control requirements.
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2.3 Brainstorming

Idea Group	Brainstormed Ideas	Purpose / Benefit
Data Management	Central employee table; spreadsheet import; Import Set and Transform Map.	Reduce manual entry and keep data structured.
Record Matching	Use Employee ID as coalesce field.	Match existing employees and reduce duplicate records.
Information Enrichment	Maintain department and location information.	Provide useful context for reporting.
Analytics & Reporting	Employee list, location report, department report, dashboard.	Make employee information easier to analyse.
Security & Access Control	HR-specific roles and ACLs.	Limit access to authorized users.
Usability / Navigation	Expose relevant ServiceNow modules.	Make the solution easier to access.

Selected solution: a ServiceNow workflow that combines spreadsheet ingestion, employee matching, reporting, and role-based access control.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Stage	Experience	Interaction
Entice	HR identifies the need to centralize employee data.	Reviews employee information and prepares the spreadsheet.
Enter	HR starts the import process.	Uploads the spreadsheet through ServiceNow Import Sets.
Engage	Records are transformed into the target table.	Runs the Transform Map and uses Employee ID for matching.
Exit	Imported records are verified.	Checks employee name, email, department, Employee ID and location.
Extend	Employee information is used for reporting.	Views the Employee Analytics Dashboard by location and department.

3.2 Solution Requirement

FR No.	Functional Requirement
FR-1	Import employee information from a spreadsheet using ServiceNow Import Sets.
FR-2	Map spreadsheet columns to the Employee Test table using a Transform Map.
FR-3	Use Employee ID as the coalesce field for existing-record matching.
FR-4	Create new employee records and update matching records through transformation.
FR-5	Display employee name, email, department, Employee ID and location.
FR-6	Provide employee analytics by department and location.

NFR No.	Non-Functional Requirement
NFR-1	Usability – the import and record-management process should be simple for HR users.
NFR-2	Security – employee information should be controlled using ServiceNow roles and ACLs.
NFR-3	Reliability – valid spreadsheet records should be processed consistently.
NFR-4	Performance – bulk data processing should reduce manual record-by-record entry.
NFR-5	Availability – authorized users should be able to access records and dashboards.
NFR-6	Scalability – additional spreadsheet records can be processed using the same workflow.

3.3 Data Flow Diagram

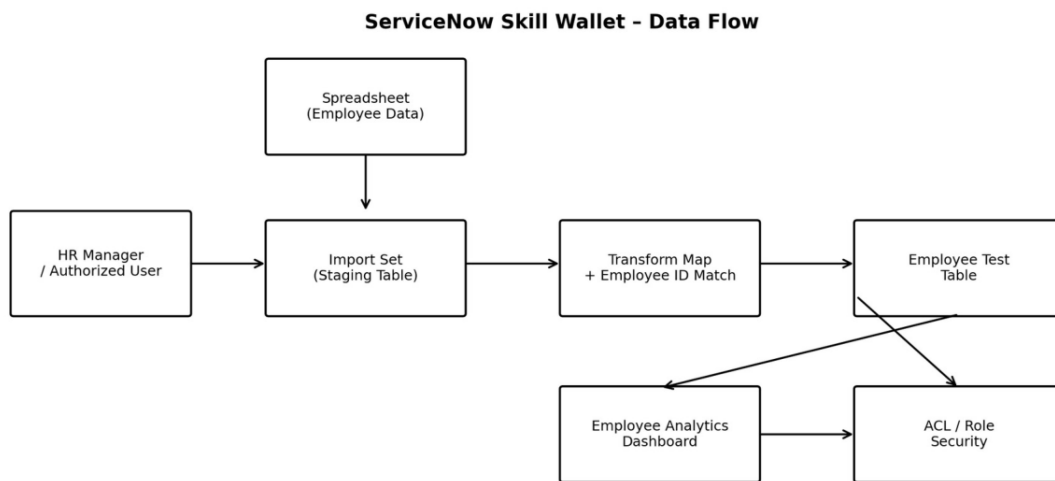


Figure 1: Data flow of the ServiceNow employee-data import solution.

Data flow: HR User / Authorized User → Spreadsheet → Import Set / Employee Import staging table → Transform Map → Employee Test table → Employee Analytics Dashboard. Security controls are applied through roles and ACLs.

3.4 Technology Stack

Layer / Component	Technology / Configuration
User Interface	ServiceNow Web UI
Import Layer	ServiceNow Import Sets and Employee Import staging table
Transformation Layer	Sample Spreadsheet Import Transform Map
Record Matching	Coalesce using Employee ID
Data Layer	Employee Test table / ServiceNow-managed platform database
Reporting	ServiceNow Reports and Employee Analytics Dashboard
Security	ServiceNow Roles and ACLs
Infrastructure	ServiceNow Cloud Platform
External APIs / ML	Not used in this project

4. PROJECT DESIGN

4.1 Problem Solution Fit

Area	Project-specific Content
Customer Segment(s)	HR personnel / HR managers maintaining employee information.
Problems	Import employee data accurately, avoid unnecessary duplicates, and make data available for reporting.
Triggers	Need for centralized employee data, reduced manual entry, record updates, and reporting.
Constraints	Consistent data structure, reliable Employee ID matching, and use within the existing ServiceNow environment.
Root Cause	Spreadsheet-based employee information requires manual handling and makes centralized reporting harder.
Solution	ServiceNow Import Set + staging table + Transform Map with Employee ID coalesce, followed by reports/dashboard.

4.2 Proposed Solution

The proposed solution is a ServiceNow-based employee data workflow. HR prepares the spreadsheet, loads it into the Employee Import staging table, and runs the Sample Spreadsheet Import Transform Map. Five source-to-target mappings are used: u_name → u_employee_name, u_email → u_email, u_department → u_department, u_employee_id → u_employee_id, and u_location → u_location. Employee ID is configured for coalesce so existing employees can be matched during re-import. The final records are displayed through reports and the Employee Analytics Dashboard.

4.3 Solution Architecture

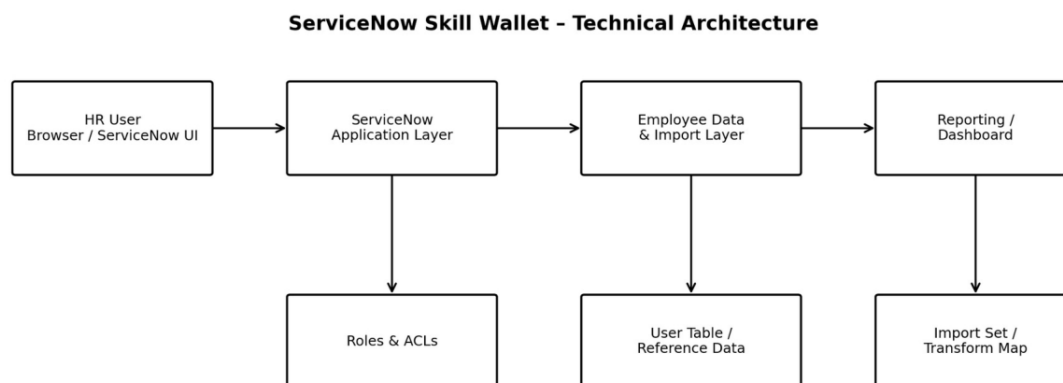


Figure 2: Technical architecture of the ServiceNow solution.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Scope	Story Points
Sprint 1	Data collection, Employee table, Import Set, Transform Map, transformation and validation.	12
Sprint 2	Coalesce/deduplication, new Excel data, reports and dashboard.	16
Total	Total completed story points.	28

Velocity = 28 story points / 2 sprints = 14 story points per sprint.

Final dataset: 17 employee records.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test ID	Scenario	Actual Result	Status
FT-01	Employee data field validation	Valid rows imported; missing name values were identified and corrected.	Pass
FT-02	Spreadsheet import	Rows loaded successfully into u_employee_import.	Pass
FT-03	Transform Map field mapping	All required source fields mapped to target fields.	Pass
FT-04	Coalesce duplicate prevention	Re-import test produced 4 updates and 0 inserts; no duplicate rows were created.	Pass
PT-01	Transform run-time test	Transform completed successfully in 0 seconds in the recorded test.	Pass
PT-02	Bulk import test	17 records processed with 0 errors.	Pass
PT-03	Identical re-import test	4 records were ignored on the recorded identical re-run; no errors.	Pass

User Acceptance Testing covered 8 test cases: 2 for data import and transformation, 2 for coalesce/deduplication, 3 for reports, and 1 for the dashboard. All 8 passed. Three identified bugs were closed: duplicate records on re-import, blank Employee Name values, and an incorrect Location chart type.

7. RESULTS

7.1 Output Screenshots

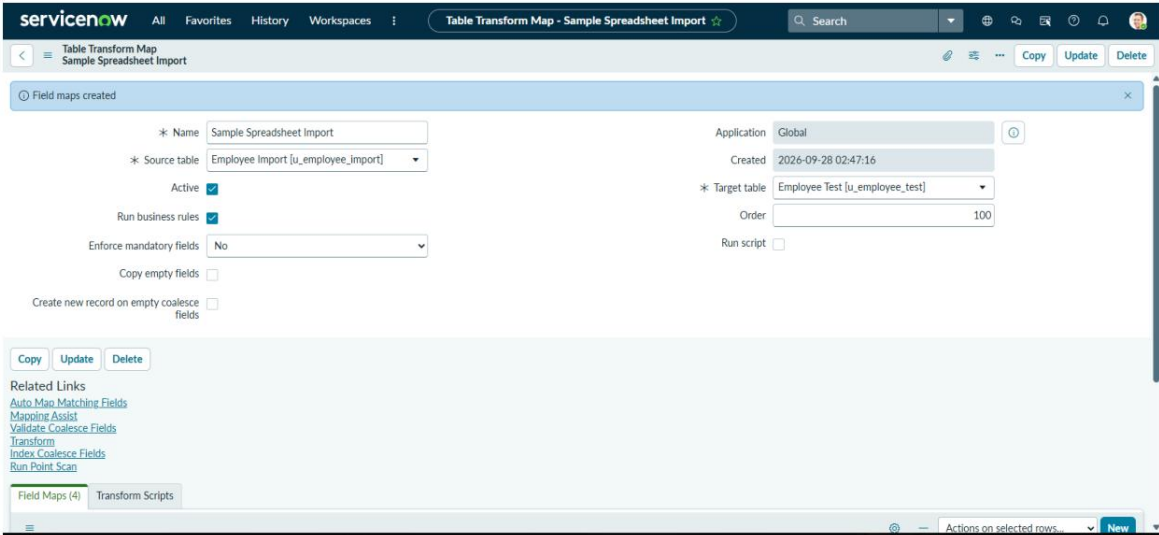


Figure 3: Sample Spreadsheet Import Transform Map configuration.

Field Maps (5) Transform Scripts			
Actions on selected rows... New			
	Source field	Target field	Coalesce
☐	u_name	u_employee_name	false
	u_email	u_email	false
	u_department	u_department	false
	u_employee_id	u_employee_id	false
	u_location	u_location	false
1 to 5 of 5			

Figure 4: Transform Map field mappings.

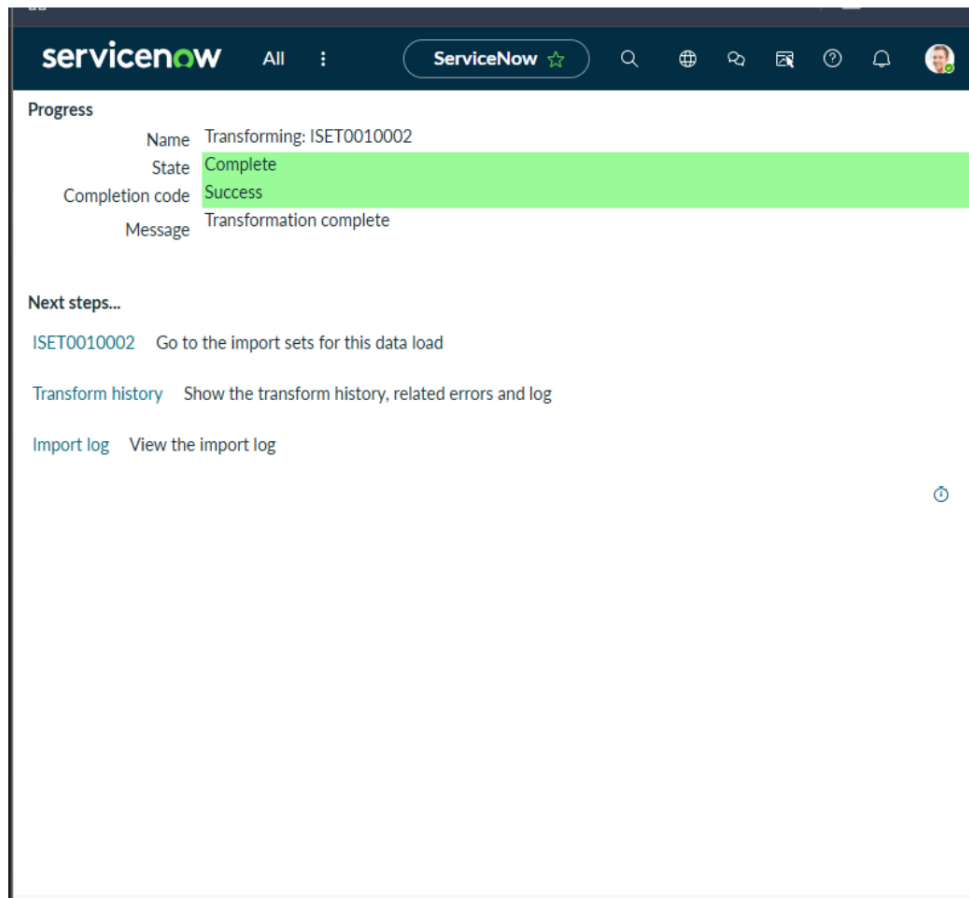


Figure 5: Successful transformation.

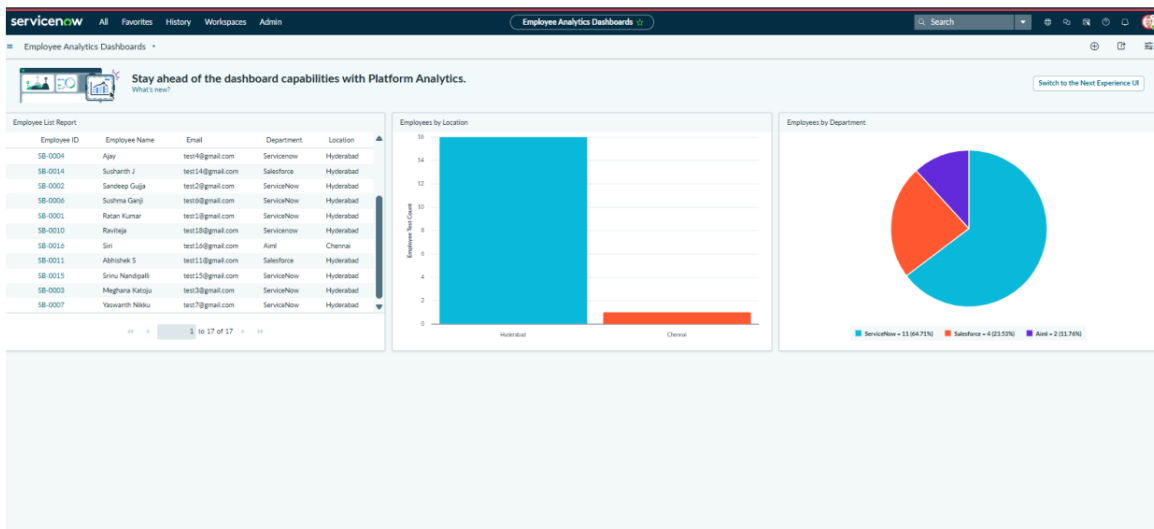


Figure 6: Employee Analytics Dashboard with list, location and department reports.

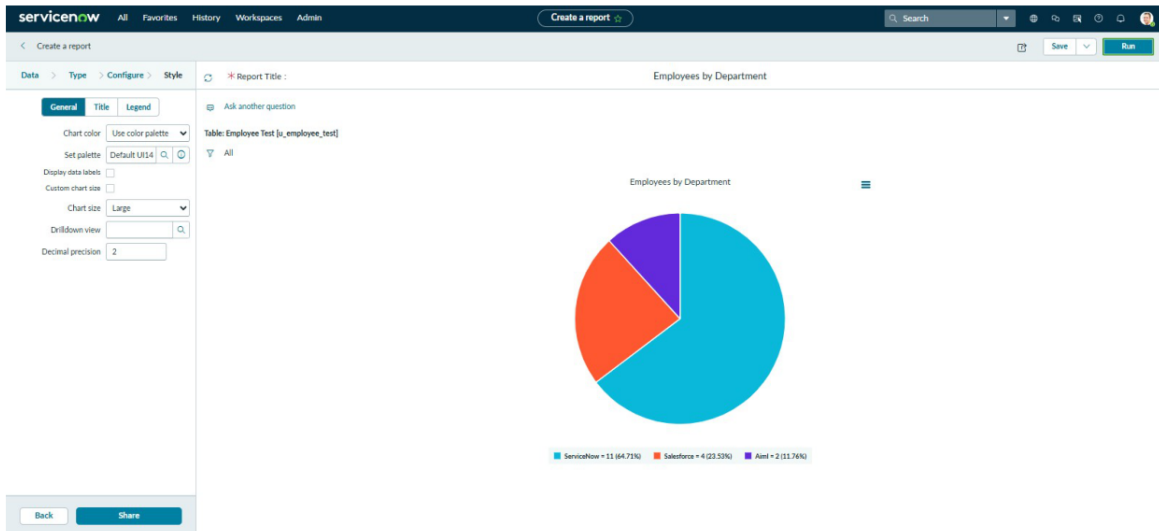


Figure 7: Employees by Department report.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Reduces repetitive manual data entry.	Depends on correctly structured spreadsheet input.
Employee ID coalesce supports reliable matching and duplicate prevention.	ServiceNow configuration and permissions are required.
Centralized employee records support reporting.	The current solution is focused on the defined employee dataset and fields.
Reports and dashboard provide quick visibility.	Changes to source columns may require mapping updates.
Roles and ACLs support controlled access.	No external public demo or standalone application is provided.

9. CONCLUSION

The project successfully demonstrates spreadsheet-based employee-data ingestion in ServiceNow using Import Sets and Transform Maps. The implementation stores structured employee records, uses Employee ID coalesce for matching, and provides reports and a dashboard for department and location analysis. Testing confirmed successful processing of the project dataset and closure of the identified defects.

10. FUTURE SCOPE

Future enhancements can include additional employee and training fields, automated scheduled imports, stronger validation rules for spreadsheet data, richer dashboard filters, audit-oriented reporting, notification workflows, and expanded role-based access for different HR responsibilities.

11. APPENDIX

Source Code: Not applicable as a standalone source-code repository; the project is implemented through ServiceNow configuration.

Dataset Link: Not applicable; the test dataset was supplied as a spreadsheet for the ServiceNow import process.

GitHub & Project Demo Link: Not applicable; the working demonstration is hosted in the ServiceNow Personal Developer Instance used for the project.

Team ID: SWTID-2026-7612

Project Name: Import Data Using Transform Maps (Spreadsheet)

PROJECT DOCUMENTATION

Import Data Using Transform Maps (Spreadsheet)

Team ID: SWTID-2026-7612

1. Introduction

Project Title: Import Data Using Transform Maps (Spreadsheet)

Team ID: SWTID-2026-7612

Team Members: Gopika Sri G K, Poojasri T S, Harini Eswari Ramesh Babu

This document describes the functional setup, architecture, operation, testing and maintenance considerations for the ServiceNow-based employee-data import solution.

2. Project Overview

Purpose

The solution imports employee information from a spreadsheet into ServiceNow, transforms the staged data into the Employee Test table, uses Employee ID for record matching, and presents the processed information through reports and an Employee Analytics Dashboard.

Features

- Spreadsheet upload through ServiceNow Import Sets.
- Employee Import staging table for source data.
- Sample Spreadsheet Import Transform Map.
- Five source-to-target field mappings.
- Employee ID coalesce for existing-record matching.
- Employee List, Employees by Location and Employees by Department reports.
- Employee Analytics Dashboard.
- Role and ACL-based access control.

3. Architecture

Frontend / User Interface

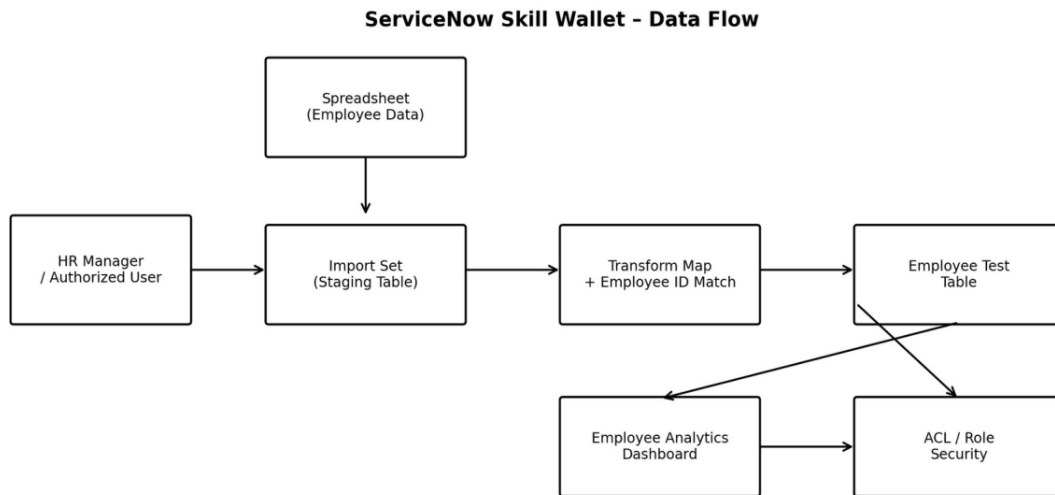
The user interface is the ServiceNow Web UI. HR or authorized users interact with Import Sets, transformation configuration, employee records, reports and the Employee Analytics Dashboard.

Backend / Application Logic

ServiceNow Import Sets handle spreadsheet ingestion and staging. The Sample Spreadsheet Import Transform Map performs field mapping and transformation. Employee ID is used as the coalesce field for matching existing records.

Database

The Employee Import table acts as the staging data store and the Employee Test table stores the transformed employee records. The platform database is managed by ServiceNow.



Architecture diagram.

4. Setup Instructions

Prerequisites

- Access to a ServiceNow Personal Developer Instance or equivalent ServiceNow environment.
- Appropriate administrative/configuration permissions for creating tables, Import Sets, Transform Maps, reports and ACLs.
- An employee spreadsheet containing the required columns.
- The required ServiceNow application/table configuration.

Installation / Configuration

1. Create or open the Employee Test table and define the required employee fields.
2. Create the Employee Import staging table and load the spreadsheet.
3. Create the Sample Spreadsheet Import Transform Map.
4. Configure the five source-to-target field mappings.
5. Set Employee ID as the coalesce field for record matching.

6. Run the transform and verify the transformation history.
7. Create the Employee List, Employees by Location and Employees by Department reports.
8. Add the reports to the Employee Analytics Dashboard.
9. Configure the required roles and ACLs for controlled access.

5. Folder Structure

Because the project is implemented inside ServiceNow rather than as a local MERN application, the logical project structure is organized by platform components:

Component	Purpose
Employee Test table (u_employee_test)	Target table containing transformed employee records.
Employee Import table (u_employee_import)	Staging table receiving spreadsheet records.
Sample Spreadsheet Import	Transform Map used for source-to-target mapping and transformation.
Reports	Employee List, Location and Department reports.
Employee Analytics Dashboard	Consolidated reporting view.
Roles & ACLs	Access-control configuration for employee information.

6. Running the Application

No local frontend/backend start commands are required. The solution runs in the ServiceNow instance.

1. Sign in to the authorized ServiceNow instance.
2. Open the Import Set / employee-data import functionality.
3. Load the employee spreadsheet into the Employee Import staging table.
4. Run the Sample Spreadsheet Import Transform Map.
5. Review transformation results and verify Employee Test records.
6. Open the Employee Analytics Dashboard to view the reports.

7. API Documentation

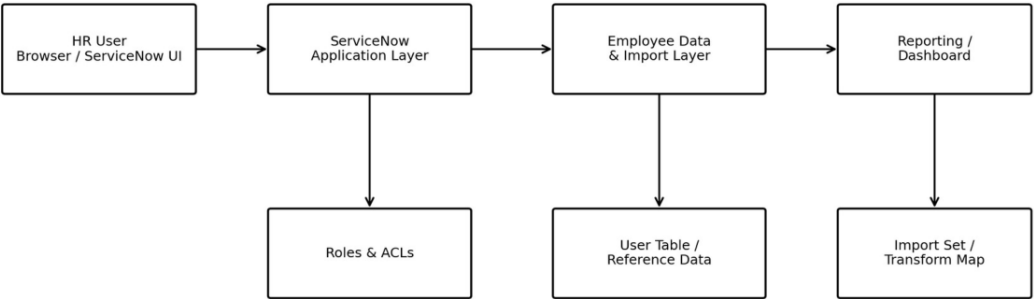
No external REST or third-party API is used by the implemented solution. Data movement is handled by ServiceNow Import Sets and Transform Maps within the platform.

8. Authentication

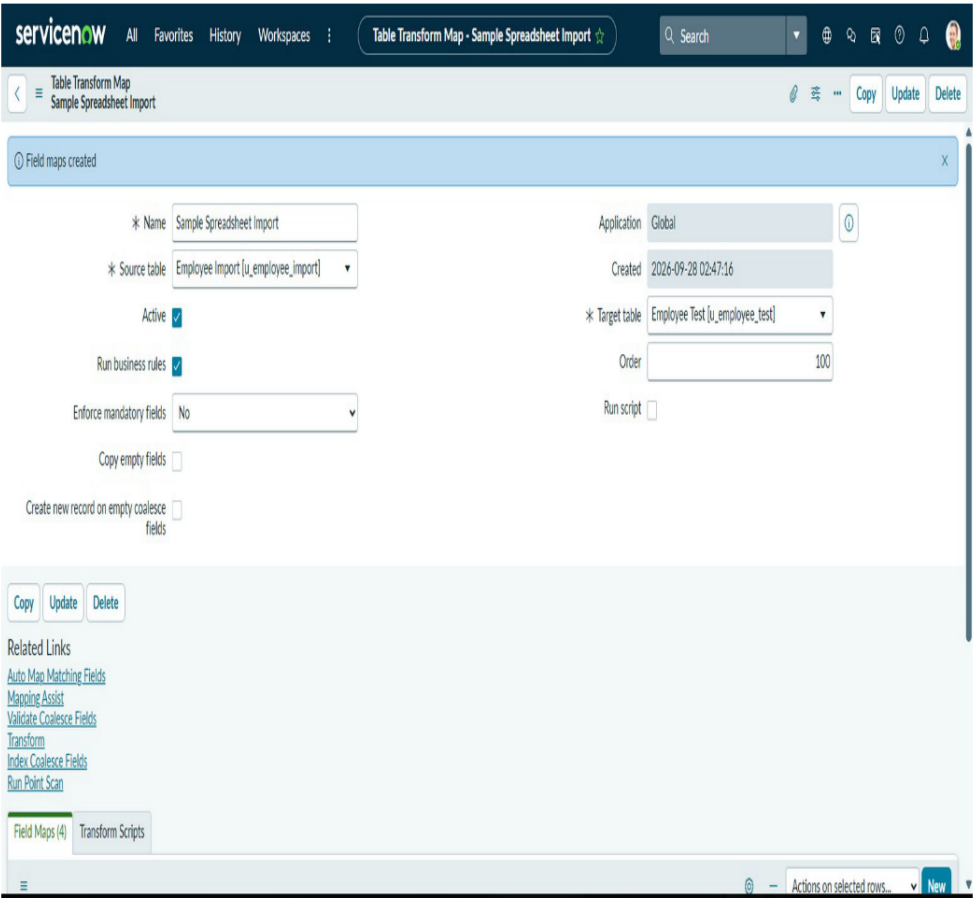
Access is provided through the ServiceNow instance's authentication mechanism. Authorization for employee-data operations is controlled through ServiceNow roles and ACLs. The project configuration includes HR-oriented access controls so that employee information is available only to authorized users.

9. User Interface

ServiceNow Skill Wallet - Technical Architecture



Field mapping interface.



Successful transformation result.

Field Maps (5) Transform Scripts			
⋮ ⚙️ — Actions on selected rows... New			
☐	Source field	Target field	Coalesce
	u_name	u_employee_name	false
	u_email	u_email	false
	u_department	u_department	false
	u_employee_id	u_employee_id	false
	u_location	u_location	false
⏪ ⏩ 1 to 5 of 5 ⏪ ⏩			

Employee Analytics Dashboard.

10. Testing

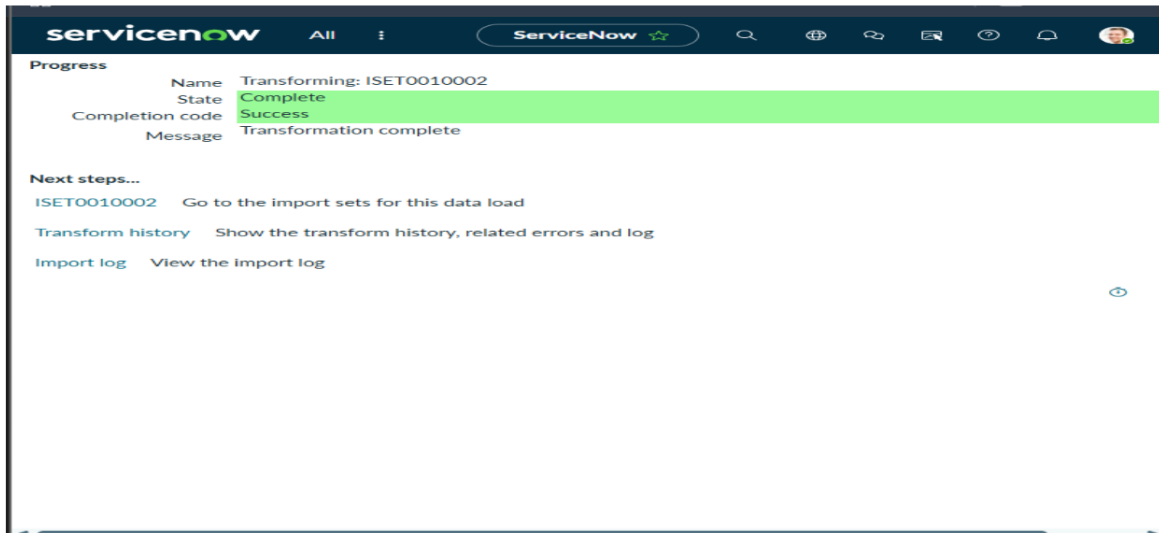
Testing covered functional field validation, spreadsheet staging, Transform Map mapping, Employee ID coalesce, bulk import, identical re-import, reports and the dashboard.

Test Area	Result
Data Import & Transform	2/2 test cases passed.
Coalesce / Deduplication	2/2 test cases passed.
Reports	3/3 test cases passed.
Dashboard	1/1 test case passed.
Total UAT	8/8 test cases passed.

Three bugs were identified and closed: duplicate records during re-import, blank Employee Name values, and an incorrect Employees by Location chart type.

11. Screenshots or Demo

The working demonstration is available in the ServiceNow Personal Developer Instance used for the project. The screenshots in this document provide evidence of the configured Transform Map, successful transformation and dashboard output.



Employees by Department report.

12. Known Issues

The solution depends on the source spreadsheet following the expected field structure. If source column names or data formats change, the Import Set and Transform Map configuration may need to be updated. The solution also requires appropriate ServiceNow permissions and a configured instance.

13. Future Enhancements

- Schedule recurring spreadsheet imports.
- Add stronger spreadsheet validation and error reporting.
- Expand the employee and training data model.
- Add dashboard filters and additional analytics.
- Introduce notification workflows for important data changes.
- Expand role-based access for different HR responsibilities.
- Add audit-oriented reports for import and record-change tracking.